3. Write the Python program for Water Jug Problem.

Aim

To write a Python program to solve the water Jug problem using the Breadth - First Search (BFS) algorithm.

Algorithm

1. Initialize two jugs with zero water
2. Define the capacities of both jugs and the target amount
3. Insert the initial state into a queue.
4. Generate possible states by filling, emptying or transferring water between jugs.
5. Avoid states that have already been visited.
6. Continue until the target amount is reached.
7. Display the steps to obtain the required quantity of water

Program

```
from collections import deque

def water_jug(a, b, target):
    queue = deque([[(0,0),[(0,0)]]])
    visited = {(0,0)}
    while queue:
```

Sample Input

Enter capacity of Jug A : 4

Enter Capacity of Jug B : 3

Enter target amount : 2

Sample Output

Solution steps:

(0, 0)
(4, 0)
(1, 3)
(1, 0)
(0, 1)
(4, 1)
(2, 3)

Result

Thus, the water Jug problem was solved Successfully using BFS.

```
(x, y), path = queue.popleft()
if x == target or y == target:
    print("Solution steps:")
    for state in path:
        print(state)
    return

states = [
    (a, y), (x, b), (0, y), (x, 0),
    (max(0, x-(b-y)), min(b, x+y)),
    (min(a, x+y), max(0, y-(a-x)))
]

for state in states:
    if state not in visited:
        visited.add(state)
        queue.append((state, path + [state]))

print("Target cannot be measured.")

a = int(input("Enter capacity of Jug A: "))
b = int(input("Enter capacity of Jug B: "))
target = int(input("Enter target amount: "))

water_jug(a, b, target)
```